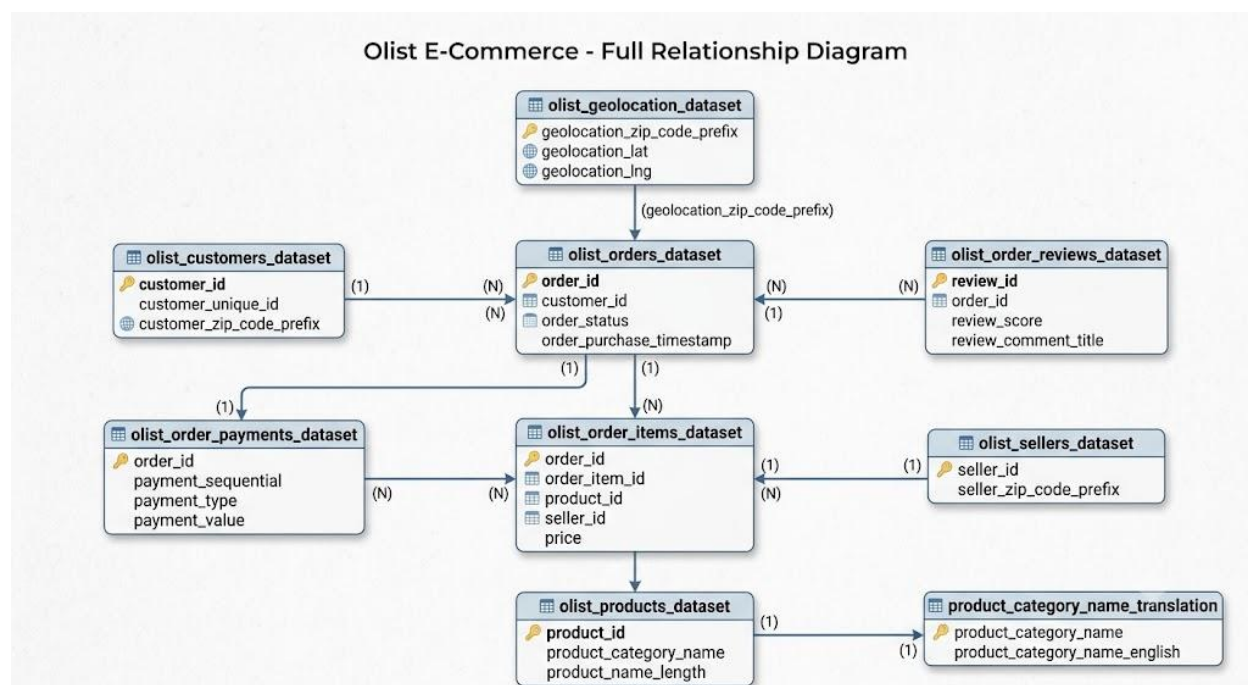


1. Data Model

Table	Description
olist_customers_dataset	Stores customer details and location info (zip code, city, state).
olist_orders_dataset	Stores core order flow, status, purchase and delivery timestamps.
olist_order_items_dataset	Stores items purchased within each order, seller ID, price, and freight.
olist_order_payments_dataset	Stores order payment details (payment type, installments, payment value).
olist_order_reviews_dataset	Stores customer review ratings and review comments for completed orders.
olist_products_dataset	Stores product attributes (category name, dimensions, weight, photo counts).
olist_sellers_dataset	Stores seller details and seller location info (zip code, city, state).
olist_geolocation_dataset	Stores Brazilian geographic coordinates (zip code prefix, latitude, longitude).
product_category_name_translation	Translates product category names from Portuguese to English.



2. Metrics Definition

Metrics	Formula	Business Purpose
Revenue	SUM(payment_value)	Measure total sales revenue
Orders	COUNT(order_item_id)	Measure transaction volume
Average Order Value	Revenue / Orders	Measure customer purchasing behavior
Top Products	Total Sales by Product	Identify best-selling products
Monthly Revenue	Revenue grouped by Month	Monitor business growth over time
Sales by Category	Revenue grouped by Product Category	Compare sales performance across product categories

3. SQL Analysis

Revenue

Purpose: Calculate the total revenue generated from all completed customer payments.

SQL: <pre>select round(sum(payment_value),2) as total_revenue from olist_order_payments_dataset;</pre>	Result: \$48,026,616.36
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Business Interpretation: Total revenue represents the overall sales performance and serves as the primary indicator of business growth.

Orders

Purpose: Calculate the total number of completed orders.

SQL: <pre>select count(order_item_id) as total_items_sold from olist_order_items_dataset;</pre>	Result: 337,950
---	-------------------------------

Business Interpretation: The total number of completed orders reflects overall transaction volume and customer purchasing activity.

Average Order Value

Purpose: Calculate the average revenue generated per completed order.

SQL:	Result:
<pre>select round(sum(p.payment_value) / count(i.order_item_id),2) as avg_order_value from olist_order_payments_dataset p join olist_order_items_dataset i on p.order_id = i.order_id;</pre>	172.69

Business Interpretation: Average Order Value helps evaluate customer purchasing behavior and overall spending patterns.

Top Products

Purpose: Identify the highest-performing products based on total sales revenue.

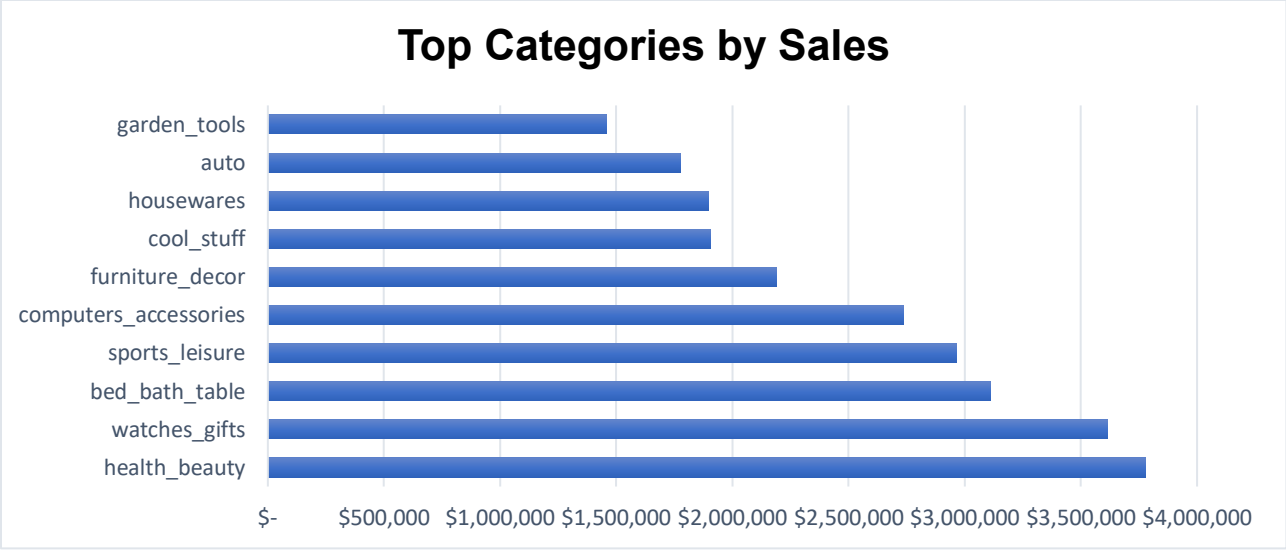
SQL:

```
select
    coalesce(t.product_category_name_english,t.product_category_name) as category_name,
    count(i.order_item_id) as total_items_sold,
    round(sum(i.price),2) as total_sales
from olist_order_items_dataset i
join olist_products_dataset p on i.product_id = p.product_id
left join product_category_name_translation t on p.product_category_name =
t.product_category_name
group by category_name
order by total_sales desc
limit 10;
```

Result:

category_name	total_items_sold	total_sales
health_beauty	29,010	3,776,044.02
watches_gifts	17,973	3,615,017.04
bed_bath_table	33,345	3,110,966.04
sports_leisure	25,923	2,964,146.91
computers_accessories	23,481	2,735,862.96
furniture_decor	25,002	2,189,287.47
cool_stuff	11,388	1,905,872.55
housewares	20,892	1,896,745.98
auto	12,705	1,778,160.33
garden_tools	13,041	1,455,769.38

Business Interpretation: Top-selling products highlight customer preferences and support inventory planning and merchandising decisions.



Monthly Revenue

Purpose: Analyze monthly revenue trends over time.

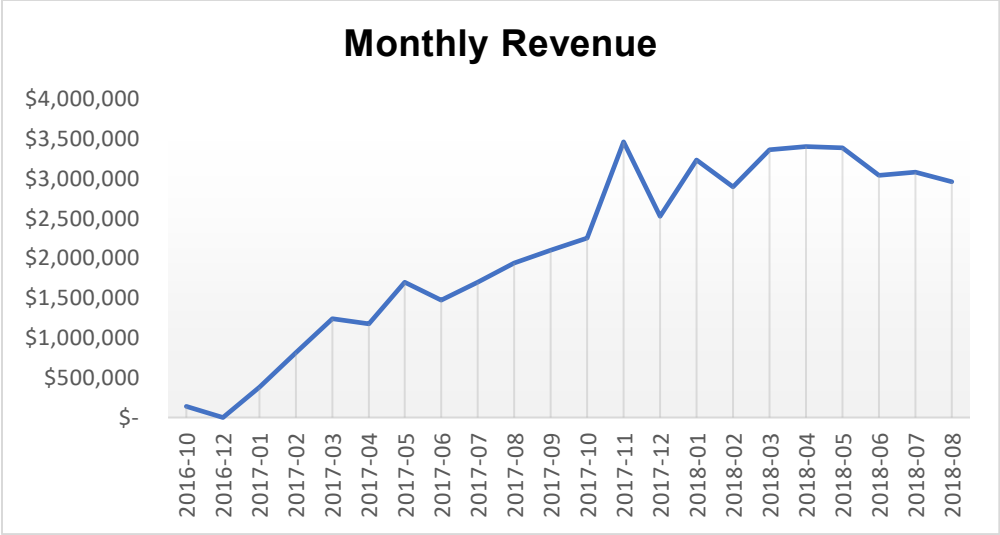
SQL:

```
select
    strftime('%Y-%m', order_purchase_timestamp) as year_month
    round(sum(p.payment_value),2) as monthly_revenue
from olist_orders_dataset o
join olist_order_payments_dataset p on o.order_id = p.order_id
where o.order_status = 'delivered'
group by year_month
order by year_month;
```

Result:

year_month	monthly_revenue	total_items_sold
2016-10	139,700.13	2,970
2016-12	58.86	9
2017-01	382,637.01	8,667
2017-02	813,895.95	17,658
2017-03	1,243,108.17	27,702
2017-04	1,172,856.54	24,363
2017-05	1,701,200.19	38,349
2017-06	1,470,676.80	33,183
2017-07	1,699,211.79	42,534
2017-08	1,938,001.83	45,477
2017-09	2,103,509.97	45,027
2017-10	2,253,420.81	49,095
2017-11	3,460,584.15	79,317
2017-12	2,529,597.51	57,960

Business Interpretation: Monthly revenue trends help identify seasonality, business growth, and potential sales fluctuations.



Sales by Category

Purpose: Compare sales performance across different product categories.

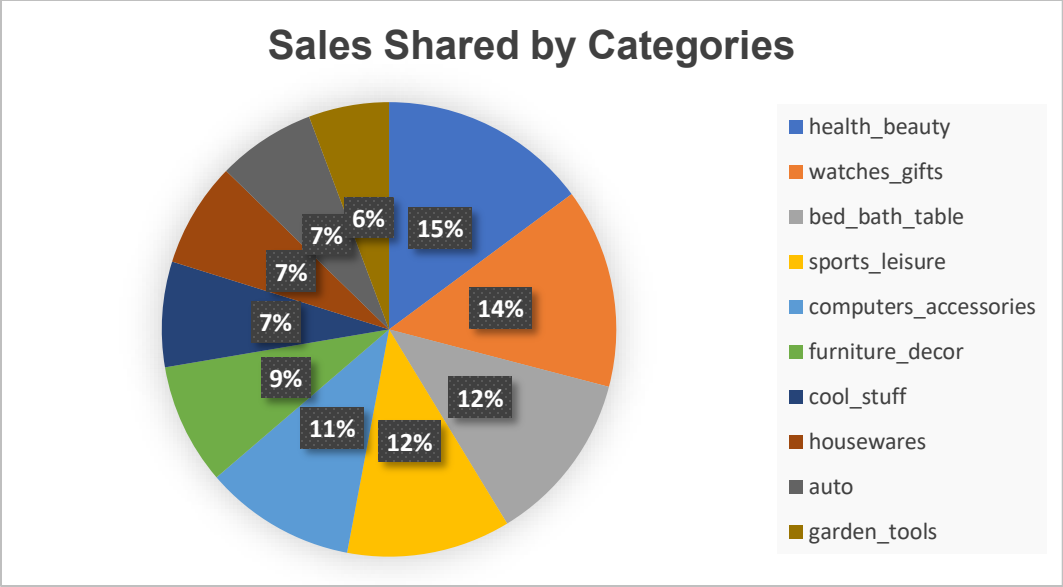
SQL:

```
select
    coalesce(t.product_category_name_english,t.product_category_name) as category_name,
    count(i.order_item_id) as total_items_sold,
    round(sum(i.price),2) as total_sales
from olist_order_items_dataset i
join olist_products_dataset p on i.product_id = p.product_id
left join product_category_name_translation t on p.product_category_name = t.product_category_name
group by category_name
order by total_sales desc
limit 10;
```

Result:

category_name	total_items_sold	total_sales
health_beauty	29,010	3,776,044.02
watches_gifts	17,973	3,615,017.04
bed_bath_table	33,345	3,110,966.04
sports_leisure	25,923	2,964,146.91
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housewares	20,892	1,896,745.98
auto	12,705	1,778,160.33
garden_tools	13,041	1,455,769.38

Business Interpretation: Category-level sales analysis helps identify high-performing product categories and supports strategic business planning.



4. Business Insights

- Monthly revenue shows a stable upward trend, indicating consistent business growth.
- A small number of products contribute a significant portion of total sales revenue, suggesting opportunities for inventory optimization and targeted marketing.
- Certain product categories consistently generate higher revenue than others and should remain strategic business priorities.
- Average order value remains relatively stable, indicating consistent customer purchasing behavior.
- The analytics dashboard enables business managers to monitor business performance in real time instead of relying on manually prepared reports, improving decision-making efficiency.

5. Recommendations

Recommendation	Business Value
Promote top-selling products	Increase Revenue
Focus marketing on high-performing categories	Improve campaign effectiveness
Monitor monthly revenue trends	Detect seasonal opportunities
Expand dashboard with customer segmentation	Improve customer insights